

SUMY, Ukraine

WMO: 332750

Lat: 50.8589N

Lon: 34.7492E

Elev: 594

StdP: 14.38

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	0.2	-10.4	3.2	-4.2	-5.2	4.3	1.0	22.9	28.7	20.8	26.6	5.4	30	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	88.1	68.3	85.0	67.0	81.9	65.8	70.7	82.7	69.2	80.8	67.7	78.8	8.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	100.5	75.9	65.2	95.4	74.0	63.8	90.7	72.6	34.9	82.5	33.6	80.9	32.4	78.9	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.3	16.5	DB	-11.3	92.5	5.3	3.8	-15.2	95.2	-18.3	97.4	-21.3	99.5	-25.1	102.3	(4)
			WB	-11.7	73.6	5.3	2.5	-15.5	75.3	-18.6	76.8	-21.6	78.2	-25.4	80.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	21.9	23.5	32.8	48.0	59.1	65.5	69.1	67.1	57.0	45.1	34.5	25.7	(6)
		DBStd	18.93	10.58	10.96	8.35	7.78	8.07	6.52	5.68	6.56	7.37	8.02	8.65	10.38	(7)
		HDD50	3725	872	741	535	128	17	0	0	20	193	466	753	(8)	
		HDD65	7382	1337	1161	1000	511	216	75	24	257	617	914	1218	(9)	
		CDD50	2229	0	0	1	68	300	464	593	531	228	42	2	0	(10)
		CDD65	410	0	0	0	1	34	89	152	118	16	0	0	0	(11)
		CDH74	3768	0	0	0	13	349	718	1302	1208	175	3	0	0	(12)
		CDH80	1103	0	0	0	1	72	161	403	433	33	0	0	0	(13)
(14)	Wind	WSAvg	8.1	9.2	9.7	9.5	8.6	7.8	7.3	6.3	6.5	7.0	7.8	8.9	9.3	(14)
(15)	Precipitation	PrecAvg	23.7	1.7	1.3	1.4	1.5	2.2	2.6	3.0	2.4	1.9	1.7	2.0	1.9	(15)
(16)		PrecMax	34.5	5.1	2.8	3.1	4.6	6.1	5.5	7.0	6.6	4.2	6.2	5.7	4.8	(16)
(17)		PrecMin	15.3	0.0	0.1	0.2	0.2	0.2	0.5	0.5	0.2	0.2	0.0	0.4	0.3	(17)
(18)		PrecStd	4.9	1.1	0.7	0.8	1.0	1.4	1.2	1.6	1.3	1.2	1.4	1.3	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	41.9	44.3	59.3	75.1	86.1	87.7	92.4	95.4	83.9	71.6	57.0	47.6	(19)	
(20)		MCWB	40.7	40.9	47.7	56.1	65.2	67.5	70.0	67.7	63.1	58.5	52.5	46.2	(20)	
(21)		DB	38.4	40.0	52.7	70.3	81.6	84.3	87.7	88.9	79.3	65.9	52.0	43.2	(21)	
(22)		MCWB	36.8	38.3	44.3	54.3	62.5	66.7	69.1	68.3	60.9	56.8	49.1	42.1	(22)	
(23)		DB	36.0	37.0	47.6	65.6	78.1	81.4	84.6	84.8	74.4	60.5	48.6	40.2	(23)	
(24)		MCWB	34.8	35.5	41.7	52.4	61.8	65.9	68.4	66.8	60.0	54.4	46.6	39.2	(24)	
(25)		DB	34.2	35.2	43.7	61.8	73.8	78.4	81.3	80.6	69.9	56.3	45.6	36.9	(25)	
(26)		MCWB	33.3	33.9	39.4	50.2	59.9	64.6	67.2	65.2	58.3	51.5	44.1	35.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	41.1	41.4	49.0	59.0	67.2	71.3	73.2	72.5	65.6	61.3	53.4	46.3	(27)	
(28)		MCDB	41.8	43.1	57.2	70.8	80.5	81.5	86.4	85.1	76.1	67.3	56.2	47.7	(28)	
(29)		WB	37.1	38.5	45.3	55.7	64.9	69.2	71.1	69.8	63.4	57.8	49.9	42.2	(29)	
(30)		MCDB	38.0	39.7	51.0	66.2	77.6	80.1	83.2	83.1	73.5	63.7	51.5	43.0	(30)	
(31)		WB	35.0	35.8	42.2	53.7	63.0	67.6	69.7	68.1	61.7	54.8	46.8	39.3	(31)	
(32)		MCDB	35.8	36.7	46.7	62.7	74.5	78.0	81.1	82.0	70.9	59.6	48.2	40.2	(32)	
(33)		WB	33.3	34.2	39.7	51.8	61.1	66.0	68.3	66.6	59.8	52.4	44.2	36.1	(33)	
(34)		MCDB	34.1	35.0	43.3	59.9	71.3	75.8	78.5	79.0	68.1	56.0	45.4	36.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.4	9.9	11.8	16.9	19.0	17.5	18.1	20.0	17.6	13.5	8.1	7.7	(35)	
(36)		5% DB	6.2	8.6	18.5	23.2	24.0	21.4	22.6	26.1	25.3	20.9	12.2	8.1	(36)	
(37)		MCWBR	6.0	7.6	11.5	11.1	9.7	8.8	8.2	9.1	11.1	12.2	9.7	7.9	(37)	
(38)		5% WB	6.0	8.1	16.6	19.7	21.1	19.0	20.0	23.2	21.2	18.0	10.4	7.9	(38)	
(39)		MCWBR	5.9	7.5	11.2	10.3	9.7	8.9	8.6	9.2	10.8	11.6	9.1	7.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.339	0.368	0.444	0.404	0.393	0.425	0.425	0.398	0.361	0.341	0.311	(40)	
(41)		taud	2.160	2.085	2.153	2.084	2.281	2.332	2.255	2.240	2.279	2.359	2.291	2.223	(41)	
(42)		Ebn at Noon	218	241	259	252	270	274	262	255	248	236	206	198	(42)	
(43)		Edh at Noon	24	34	38	46	40	38	41	39	34	26	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	295	590	983	1346	1810	1927	1843	1630	1075	595	252	180	(44)	
(45)		RadStd	55	89	164	123	157	169	112	144	136	94	54	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.31	N/A	N/A	+1.60	N/A	N/A	-239	-365	+212	+106
Regional (7 neighbors)		+1.39	N/A	N/A	+1.67	N/A	N/A	-281	-394	+220	+112

Nomenclature: See separate page